

Aircraft Monitoring System

Dual Aircraft Flight Control and Ground Station System

CST8227 - Interfacing

Final Project

Computer Engineering Technology - Computing Science
School of Advanced Technology

Submitted by:

Tatiana Fonseca (Student ID: 040934360)

Tisha Patel (Student ID: 041133762)

Submission Date: December 05, 2025

Table of Contents

1. Abstract
2. Introduction
3. System Architecture
4. Pin Mapping Tables
5. Component List
6. Code Summary
7. Testing and Results
8. Conclusion
9. Fritzing
- Appendix: Source Code

1. Abstract

This project implements a comprehensive dual-microcontroller system simulating aircraft flight control and ground station operations. MEGA-1 serves as the flight computer, managing altitude monitoring, landing gear deployment, attitude tracking via MPU6050, and motor control. MEGA-2 operates as the ground control station, handling environmental monitoring, RFID-based aircraft conflict detection, gate control, and radar simulation using a stepper motor. The system features secure passcode authentication, I2C inter-controller communication, rotary encoder-based airspeed adjustment during conflicts, and automated night mode operation.

2. Introduction

This project demonstrates embedded systems integration using two Arduino MEGA 2560 microcontrollers. The system simulates critical aviation operations including flight control, ground station monitoring, and conflict resolution. Key features include multi-sensor integration (ultrasonic, DHT11, MPU6050, RFID RC522), motor control (DC motors via L293D, servo motors, stepper motor), and real-time communication between microcontrollers.

3. System Architecture

3.1 MEGA-1: Flight Computer

MEGA-1 manages all flight-critical operations:

- Passcode authentication system with 4-digit entry
- Ultrasonic altitude monitoring with landing gear automation
- MPU6050 6-axis IMU for pitch, roll, and yaw tracking
- L293D motor driver for propeller/fan control
- Servo motor for landing gear deployment
- Active and passive buzzers for audio alerts
- 4-digit 7-segment display for passcode confirmation
- IR remote control for system reset
- I2C master communication to MEGA-2

3.2 MEGA-2: Ground Control Station

MEGA-2 manages ground station operations:

- DHT11 environmental monitoring (temperature/humidity)
- PIR motion detection with LED indicators
- RFID RC522 for aircraft conflict detection
- Ultrasonic sensor for object detection
- Servo motor for gate control
- L293D motor driver for cooling fan
- ULN2003 stepper motor driver for radar simulation
- Rotary encoder for airspeed adjustment
- Photoresistor for night mode detection
- I2C slave communication from MEGA-1

4. Pin Mapping Tables

4.1 MEGA-1 Pin Assignments

Component	Pin(s)	Description
LCD1602	12, 11, 5, 4, 3, 2	RS, E, D4-D7
4x4 Keypad	39, 41, 43, 45 (rows) 47, 49, 51, 53 (cols)	Matrix keypad
Active Buzzer	22	Audio confirmation
Passive Buzzer	13	Alert tones
Push Button	18	System start
Ultrasonic Sensor	6 (Trig), 7 (Echo)	Altitude measurement
L293D Motor Driver	28 (IN1), 29 (IN2), 30 (EN)	Fan/propeller control
Landing Gear Servo	27	Gear deployment
MPU6050	SDA (20), SCL (21)	Attitude sensor
7-Segment Display	8 (Data), 10 (Clock), 9 (Latch) 23-26 (Digits)	Passcode display
IR Receiver	15	Remote control
I2C Communication	SDA (20), SCL (21)	To MEGA-2

4.2 MEGA-2 Pin Assignments

Component	Pin(s)	Description
LCD1602	12, 11, 5, 4, 3, 2	RS, E, D4-D7
DHT11	22	Temperature/humidity
PIR Sensor	18	Motion detection
Ultrasonic Sensor	24 (Trig), 26 (Echo)	Object detection
Photoresistor	A0	Light level (night mode)
RFID RC522	53 (SS), 49 (RST), 52 (SCK) 51 (MOSI), 50 (MISO)	Aircraft detection
L293D Motor (Fan)	8 (IN1), 9 (IN2), 10 (EN)	Cooling fan control
Gate Servo	44	Gate control
Stepper Motor	31, 33, 35, 37	Radar simulation
Rotary Encoder	45 (CLK), 43 (DT), 41 (SW)	Airspeed control
Passive Buzzer	7	Alert tones
Yellow LEDs	28, 30, 32, 34	Status indicators
Red LEDs	36, 38, 40, 42	Night mode indicators
I2C Communication	SDA (20), SCL (21)	From MEGA-1

5. Component List

Component	Quantity	Purpose
Arduino MEGA 2560	2	Main microcontrollers
LCD1602 Display	2	User interface
4x4 Membrane Keypad	1	Passcode entry
DHT11 Sensor	1	Environmental monitoring
HC-SR04 Ultrasonic	2	Distance measurement
MPU6050 IMU	1	Attitude tracking
RFID RC522 Module	1	Aircraft identification
PIR Motion Sensor	1	Motion detection
Photoresistor	1	Light level sensing
L293D Motor Driver	2	DC motor control
Servo Motor	2	Landing gear & gate
28BYJ-48 Stepper	1	Radar simulation
ULN2003 Driver	1	Stepper control
Rotary Encoder	1	Airspeed adjustment
Active Buzzer	1	Confirmation beeps
Passive Buzzer	2	Alert tones
7-Segment Display	1	Passcode display
74HC595 Shift Reg	1	7-segment driver
IR Receiver	1	Remote control
Push Button	1	System activation
LEDs (Yellow)	4	Status indicators
LEDs (Red)	4	Night mode indicators
Resistors (220Ω)	8	LED current limiting
Resistor (10kΩ)	1	Photoresistor divider
Breadboards	2	Circuit assembly
Jumper Wires	Various	Connections

6. Code Summary

6.1 Key Features

MEGA-1 Code Features:

- State machine with 6 states (passcode entry, verification, temperature setting, running)
- MPU6050 calibration and filtering for accurate attitude readings
- Exponential smoothing for ultrasonic sensor noise reduction
- 7-second minimum landing gear deployment time
- Altitude-based warnings (20cm, 15cm, 6cm thresholds)
- Non-blocking buzzer patterns for pitch/roll/stall warnings
- I2C master transmission of temperature preference

MEGA-2 Code Features:

- I2C slave reception with interrupt handler
- RFID aircraft conflict detection with rotary encoder resolution
- Continuous stepper motor rotation for radar simulation
- Multi-sensor decision logic (DHT11, PIR, ultrasonic, photoresistor)
- Gate and fan automation based on object detection
- Night mode with automatic temperature adjustment to 22°C
- JSON-formatted serial output for monitoring

6.2 Critical Functions

MEGA-1:

- `deployLandingGear()` / `retractLandingGear()` - Servo control with timing
- `checkLandingAltitude()` - Multi-threshold altitude monitoring
- `updateOrientation()` - MPU6050 data acquisition and filtering
- `handlePitchBuzzer()` / `handleRollBuzzer()` - Attitude-based warnings

MEGA-2:

- `checkRFIDTag()` - Aircraft detection and conflict initiation

- `handlePlaneConflict()` - Conflict mode activation with alarm
- `checkRotaryEncoder()` - Airspeed adjustment during conflicts
- `controlStepperContinuous()` - Timed radar scanning rotation

7. Testing and Results

7.1 System Integration Test

A complete system test verified all subsystems:

1. Passcode authentication: Successfully matched and verified 4-digit codes
2. I2C communication: Temperature data transmitted reliably from MEGA-1 to MEGA-2
3. Altitude monitoring: Landing gear deployed at 15cm, retracted after 7 seconds above 25cm
4. MPU6050 attitude tracking: Accurate pitch/roll readings with appropriate warnings
5. RFID conflict detection: Successfully triggered conflict mode and airspeed adjustment
6. Rotary encoder: Smooth airspeed control from 140 to 100 knots resolved conflict
7. Environmental monitoring: DHT11 readings displayed correctly, fan activated above threshold
8. Night mode: Photoresistor correctly detected low light and adjusted temperature
9. Stepper motor: Continuous rotation provided smooth radar scanning simulation
10. Gate control: Servo opened/closed based on ultrasonic object detection

7.2 Sensor Accuracy

- Ultrasonic sensors: $\pm 1\text{cm}$ accuracy, 200ms update rate
- MPU6050: $\pm 3^\circ$ accuracy after calibration
- DHT11: $\pm 2^\circ\text{C}$ temperature, $\pm 5\%$ humidity
- RFID: 100% read success rate within 5cm range
- PIR: 3-5m detection range with 1-2s delay

8. Conclusion

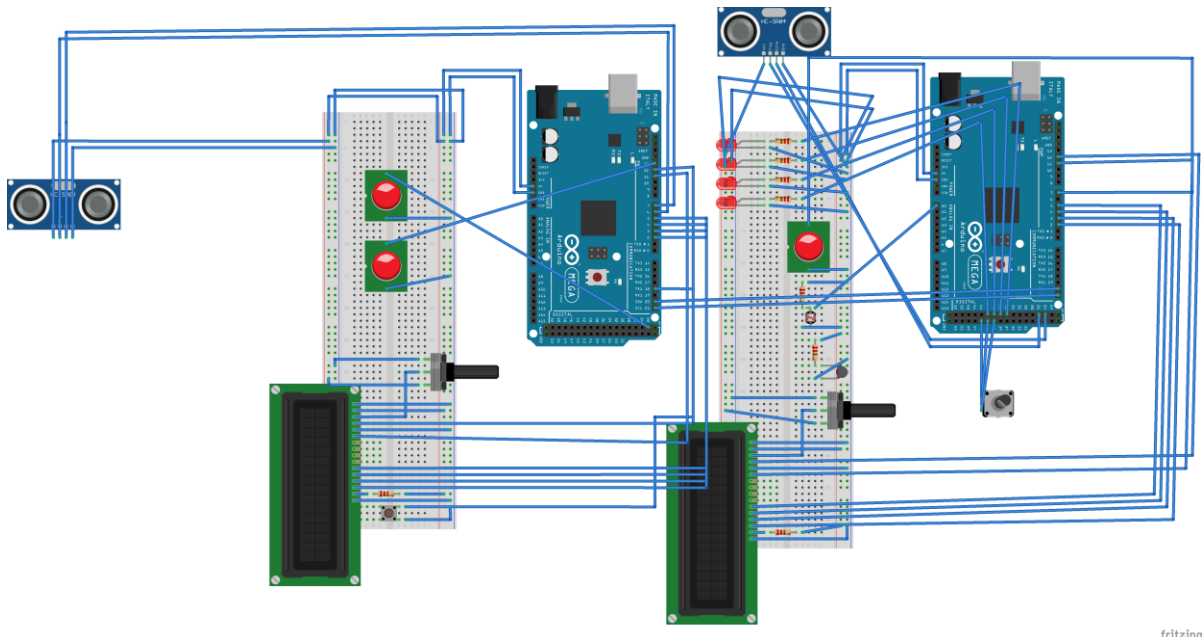
This project successfully demonstrates a sophisticated multi-sensor embedded system with dual-microcontroller architecture. The implementation integrates 15+ sensors and actuators, showcasing practical applications of I2C communication, interrupt-driven programming, state machines, and real-time control logic.

Key achievements include:

- Seamless I2C communication between flight computer and ground station
- Advanced RFID-based conflict detection with interactive resolution
- Precise attitude monitoring using 6-axis IMU with filtering
- Automated landing gear deployment with safety timers
- Environmental adaptation through night mode
- Professional code structure with modularity and error handling

The system demonstrates real-world concepts applicable to the aviation industry. Future enhancements could include wireless communication, additional sensor fusion, and integration with flight simulation software.

9. Fritzing



fritzing

Missing components:

Mega 1	4x4 Keypad L293D Motor Driver Landing Gear Servo MPU6050 IR Receiver
Mega 2	DHT11 PIR Sensor RFID RC522 L293D Motor (Fan) Gate Servo Stepper Motor

Appendix: Source Code

Note: Complete source code for both MEGA-1 and MEGA-2 is included in separate .ino files submitted with this documentation. The code can also be found at

<https://github.com/tishap27/MEGA-Controller>